Fuzzy Cache Lookups for Partially Stateful Tool Calls

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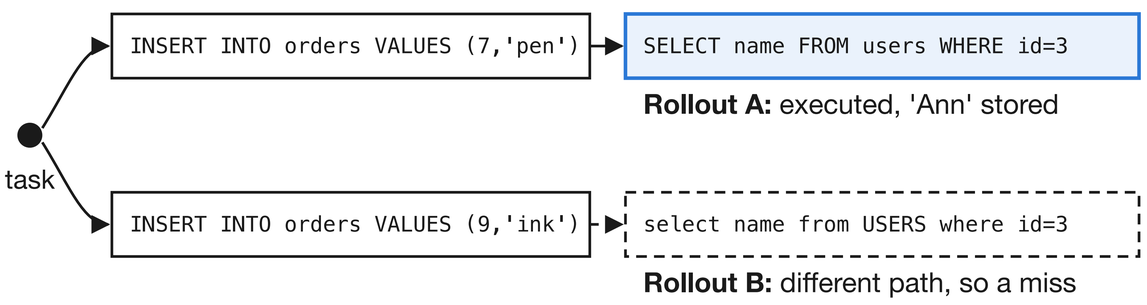
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If a cache looks at what each tool call actually *reads*, it can serve **41% of SQL calls** from cache instead of TVCache's 6%, with **no wrong results in 681 checked reuses**.

Motivation

Agentic RL samples many rollouts per task, and they repeat tool calls. TVCache [1] reuses a result only if the call *and every earlier state-changing call* match exactly, so it misses respelled queries and reads after unrelated writes.

TVCache: tree of exact call strings



Ours: checked after an exact miss



Figure 1. TVCache keys a result on the full path of state-changing calls, so rollout B misses. We key a read on its canonical text and the versions of the state it reads, so A and B share one entry.

Approach

1. **Canonicalize** each call: SQL with SQLGlot [2]; `cat`, `ls`, `echo`, `cd` with absolute paths.
2. **Find the state it reads and writes**: tables, files, the working directory.
3. **Version each piece of state** by hashing the writes to it.
4. **Exact lookup first**; on a miss, key a read by its canonical text and the versions of the state it reads.

Writes always execute; only reads are reused. Anything we can't analyze (pipes, `random()`, unknown commands) falls back to plain TVCache.

Evaluation setup

1,297 generated SQL and shell calls; every reused value is checked against real execution. *Oracle*: a perfect canonicalizer. *Hints*: read-only calls declared to TVCache.

Next steps

- Terminal-bench rollouts on a Docker host, with per-command output.
- Wider shell coverage and a state key for `sqlite_master`.
- RL runs with verl; contribute the lookup to TVCache.

Results

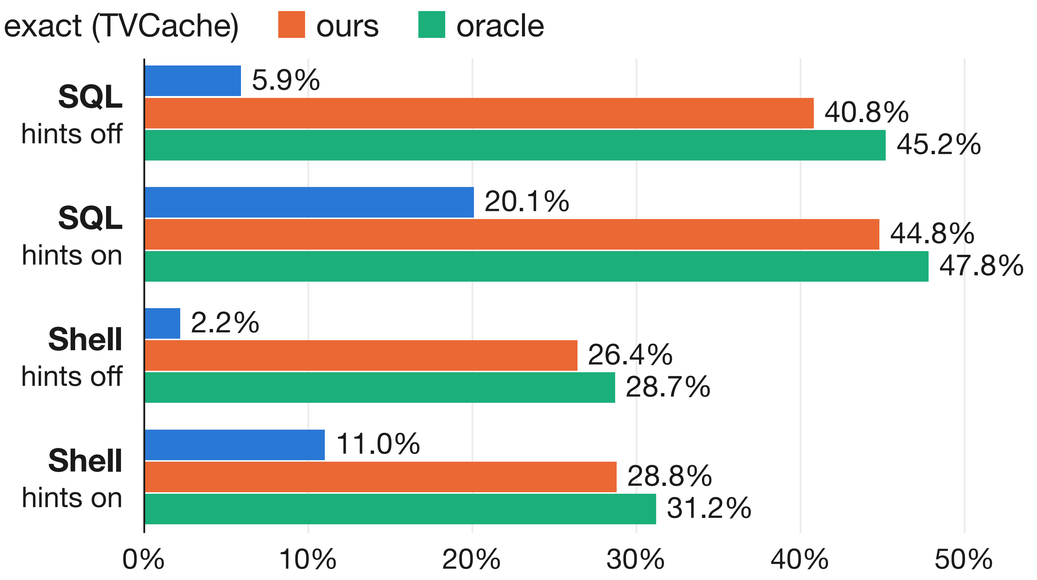


Figure 2. Share of calls served from cache. Ours reaches about 90% of the oracle.

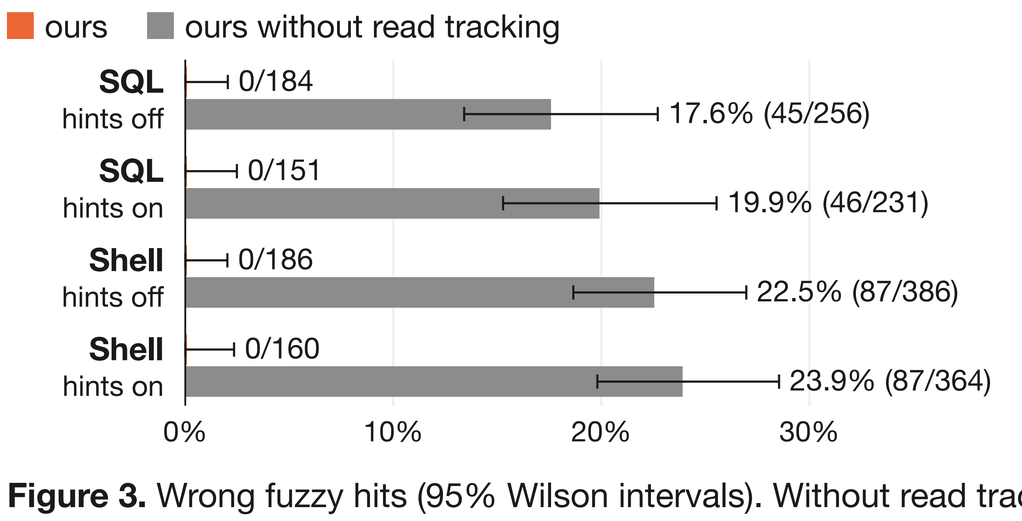


Figure 3. Wrong fuzzy hits (95% Wilson intervals). Without read tracking, 18–24% are wrong.

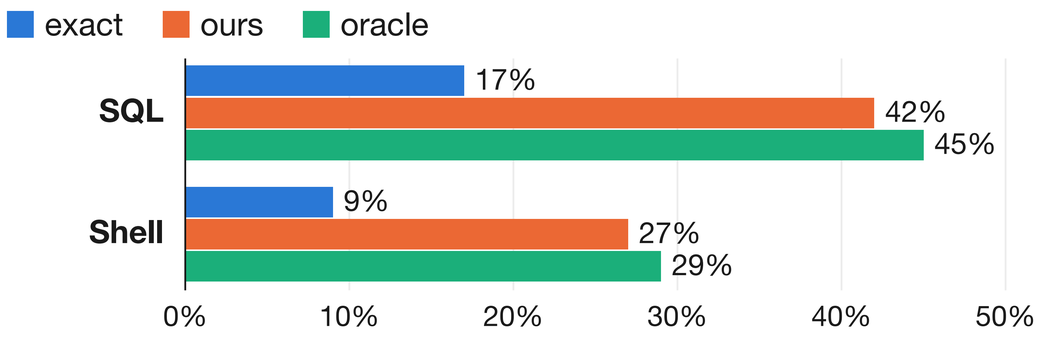


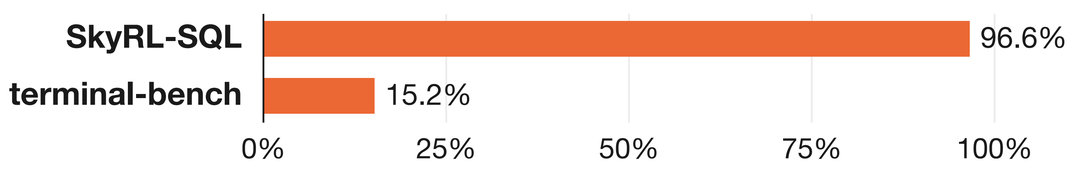
Figure 4. Executions saved vs. no cache, with hints.

Real rollouts: SkyRL-SQL

Qwen2.5-Coder-7B, 20 questions × 5 samples [3] (pilot):

mode	hit rate	wrong hits	executions
exact (TVCache)	33.2%	–	0.67
ours	34.4%	0 / 5	0.66

TVCache's 33.2% matches the 33% in [1]. These queries never write, so only spelling can add hits.



Share of reference calls our canonicalizers accept.

1. TVCache: tool-call caching for agentic RL. arXiv 2602.10986.
2. SQLGlot. github.com/tobymao/sqlglot
3. NovaSky-AI. SkyRL-SQL-653 dataset and environment.
4. Terminal-bench, via B. Kataria's container-manager.

Code: github.com/Harshaan-Chugh/fuzzy-tvcache
Compute: NERSC Perlmutter